

Title – “Introduction to Computational Thinking and Basics of Artificial Intelligence using MATLAB and Simulink”

Title/Learning Outcomes	EC	EE	CSE/IT
Common course – 30 hrs.			
Basics of MATLAB (2 hr.)	MATLAB Onramp Self-Paced Online Courses - MATLAB & Simulink – 2 hr.		
Programming in MATLAB (3.5 hrs.)	MATLAB Desktop Tools and Troubleshooting Scripts Self-Paced Online Courses - MATLAB & Simulink – 1 hr.		
	Programming Constructs Self-Paced Online Courses - MATLAB & Simulink – 1.5 hrs.		
	The How and Why of Writing Functions Self-Paced Online Courses - MATLAB & Simulink – 1 hr.		
Visualization in MATLAB (4 hrs.)	Explore Data with MATLAB Plots Self-Paced Online Courses - MATLAB & Simulink		
	Plot Beyond the Second Dimension Self-Paced Online Courses - MATLAB & Simulink – 1.5 hrs.		
	How MATLAB Graphics Work Self-Paced Online Courses - MATLAB & Simulink – 1 hr.		
Data Analysis in MATLAB (6.5 hrs.)	Tables Self-Paced Online Courses - MATLAB & Simulink – 2 hr.		
	Clean and Prepare Data for Analysis Self-Paced Online Courses - MATLAB & Simulink – 1.5 hr.		
	Common Data Analysis Techniques Self-Paced Online Courses - MATLAB & Simulink – 1 hr.		
	Find and Extract Subsets of Data Self-Paced Online Courses - MATLAB & Simulink – 1 hr.		
	Calculations on Grouped Data Self-Paced Online Courses - MATLAB & Simulink – 1 hr.		
MATLAB Skills for Simulink Modeling – 4hrs.	Make and Manipulate Matrices Self-Paced Online Courses - MATLAB & Simulink – 1.5 hrs.		
	Programming Constructs Self-Paced Online Courses - MATLAB & Simulink – 1.5 hrs.		
	The How and Why of Writing Functions Self-Paced Online Courses - MATLAB & Simulink – 1 hr.		
Simulink Basics (3.5 hrs.)	Simulink Onramp Self-Paced Online Courses - MATLAB & Simulink – 2hrs.		
	Circuit Simulation Onramp Self-Paced Online Courses - MATLAB & Simulink – 1.5 hrs.		
Theme – AI related courses (7.5 hrs.)	Statistics Onramp Self-Paced Online Courses - MATLAB & Simulink – 1hr.		
	Curve Fitting Onramp Self-Paced Online Courses - MATLAB & Simulink – 1hr.		
	Optimization Onramp Self-Paced Online Courses - MATLAB & Simulink – 1 hr.		
	Signal Processing Onramp Self-Paced Online Courses - MATLAB & Simulink – 1hr.		

	Feature Extraction Techniques for Signals Self-Paced Online Courses - MATLAB & Simulink – 1 hr.
	Machine Learning Onramp Self-Paced Online Courses - MATLAB & Simulink – 1 hr.
	Deep Learning Onramp Self-Paced Online Courses - MATLAB & Simulink – 1.5 hrs.